

- SWL-11-19

Roll No. _____ Annual 2019

Mathematics

(INTERMEDIATE PART - I, Class 11th) - (III)

Paper : I

Time: 30 Minutes

Objective

Code : 6195

Marks : 20

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question.

1. If ω is cube root of unity, then $(1 + \omega - \omega^2)^3 =$
 (A) -8ω (B) 8ω (C) 8 (D) -8
2. $\frac{p(x)}{x^3 + 1}$ will be proper fraction if degree of $p(x)$ is
 (A) 1 (B) 2 (C) 3 (D) 4
3. The series $a + ar + ar^2 + \dots$ converges if
 (A) $|r| > 1$ (B) $|r| \geq 1$ (C) $|r| \leq 1$ (D) $|r| < 1$
4. A, G, H, are in
 (A) A.P (B) G.P (C) H.P (D) series
5. For an event A, range of its probability $P(A)$ is
 (A) $-1 \leq P(A) \leq 1$ (B) $0 < P(A) < 1$ (C) $0 \leq P(A) \leq 1$ (D) $P(A) = 1$
6. If ${}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 0$, then $n =$
 (A) 0 (B) 4 (C) 6 (D) 9
7. $(1+i)^8 =$
 (A) 2 (B) 4 (C) 8 (D) 16
8. The conjunction of two statements p and q is denoted by
 (A) $p \leftrightarrow q$ (B) $p \rightarrow q$ (C) $q \rightarrow p$ (D) $p \wedge q$
9. If $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 2$ then $\begin{vmatrix} c & d \\ a & b \end{vmatrix} =$
 (A) 2 (B) -2 (C) ± 2 (D) 0
10. If $|A| = 5$, then $|A'| =$
 (A) -5 (B) $\frac{1}{5}$ (C) 0 (D) 5
11. No. of roots of the equation $(x-4)^2 = x^2 - 8x + 16$ are
 (A) 2 (B) 4 (C) 8 (D) infinite

(Turn Over)

12. In an equilateral $\triangle ABC$

- (A) $r_1 > r_2$ (B) $r_1 < r_2$ (C) $r_1 = r_2$ (D) $r_1 \neq r_2$

13. With usual notations $a + b - c =$

- (A) $2s$ (B) $2s - 2c$ (C) $2s - 2b$ (D) $2s - 2a$

14. $\sin^{-1}(0) + \cos^{-1}(0) =$

- (A) $\frac{\pi}{2}$ (B) $\frac{\pi}{6}$ (C) $\frac{2\pi}{5}$ (D) $\frac{\pi}{4}$

15. If $\sin x = -\frac{\sqrt{3}}{2}$, then $x =$

- (A) $\frac{\pi}{3}$ (B) $\frac{4\pi}{3}$ (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{3}$

16. If n is even, middle term of $(a+b)^n$ is

- (A) $(\frac{n}{2})\text{th term}$ (B) $(\frac{n+2}{2})\text{th term}$ (C) $(\frac{n+1}{2})\text{th term}$ (D) $(\frac{n+3}{2})\text{th term}$

17. 2^{nd} term of $(a+b)^7$ is

- (A) a^7 (B) $7ab^6$ (C) $7a^6b$ (D) $7ab$

18. $\tan(\alpha - 90^\circ) =$

- (A) $\cot \alpha$ (B) $-\cot \alpha$ (C) $\tan \alpha$ (D) $-\tan \alpha$

19. $\frac{\pi}{3} \text{ rad}$ is an angle.

- (A) acute (B) obtuse (C) straight (D) reflexive

20. Period of $\frac{1}{2} \sin 2x$ is

- (A) $\frac{\pi}{2}$ (B) π (C) 2π (D) 4π